



JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE
NAME

CLASS

EXAM INDEX
NUMBER

CHEMISTRY

8872/02

Paper 2 Structured Questions

14 September 2011

2 hours

Candidates answer on Question Paper.

Additional Materials: Answer Paper
 Graph Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

A *Data Booklet* is provided. Do not write anything on the *Data Booklet*.

You are reminded of the need for good English and clear presentation in your answers.

Section A

Answer **all** questions.

Section B

Answer **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together behind this question paper.
The number of marks is given in brackets [] at the end of each question or part question.

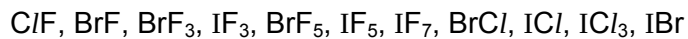
For Examiner's Use	
Section A	
B4	
B5	
B6	
Total	

This document consists of **15** printed pages and **1** blank page.

Section A

Answer **all** questions in this section in the spaces provided.

1. Some inter-halogen compounds are listed below:



- (a) What is the oxidation state of chlorine in ClF and ICl ?

oxidation state of Cl in ClF :

oxidation state of Cl in ICl :

[2]

- (b) Which diatomic compound in the list above would you expect to have the strongest bonds? Explain your answer.

.....
.....
.....

[2]

- (c) Draw diagrams to illustrate the shapes of BrF_3 and BrF_5 . Name the shape of each molecule and deduce whether the molecules are polar or non-polar.

	BrF_3	BrF_5
Shape		
Name of shape		
Polar/ Non-polar		

[6]

1. (d) The halogen, chlorine, is commonly used as a reagent in the form of chlorine gas in reactions.

(i) Describe, with an equation, the reaction between phosphorus and excess chlorine gas.

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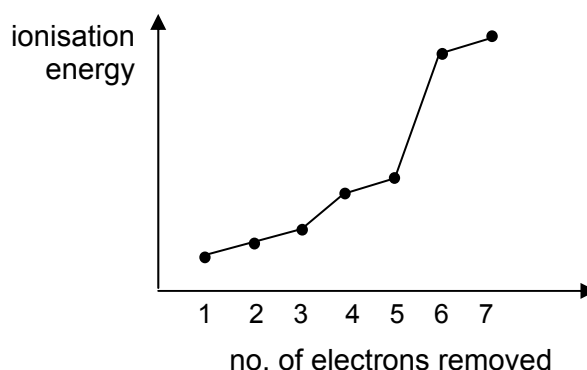
(ii) Predict, and explain in terms of the structure, the pH of the resulting solution when your product in (d)(i) is dissolved in water.

.....
.....
.....
.....

[4]

[Total: 14]

2. The figure below shows the energy required to remove seven electrons, one at a time, from an atom of element **A**, found in Period 2.



- (a) Deduce the group that element **A** belongs to in the Periodic Table. Explain your answer.

.....

.....

.....

.....

[2]

- (b) State the identity of element **A** and hence write its electronic configuration using the electrons-in-boxes configuration.

[1]

- (c) Write equations to illustrate the first and second ionisation energies of **A**.

first ionisation energy of **A** :

second ionisation energy of **A** :

[2]

- (d) Why is the energy difference between the removal of the 5th and 6th electrons much larger than the energy difference between the removal of the 4th and 5th electrons?

.....

.....

.....

[1]

2. (e) Magnesium reacts with **A** to form an ionic compound of the formula Mg_3A_2 .

(i) Draw a dot-and-cross diagram to illustrate the bonding in Mg_3A_2 .

(ii) State and explain which one of the ions in Mg_3A_2 would have a smaller ionic radius.

.....

.....

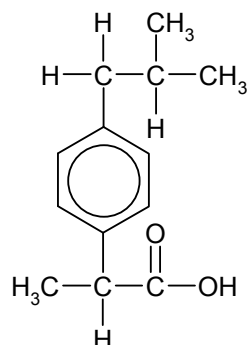
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[3]

[Total: 9]

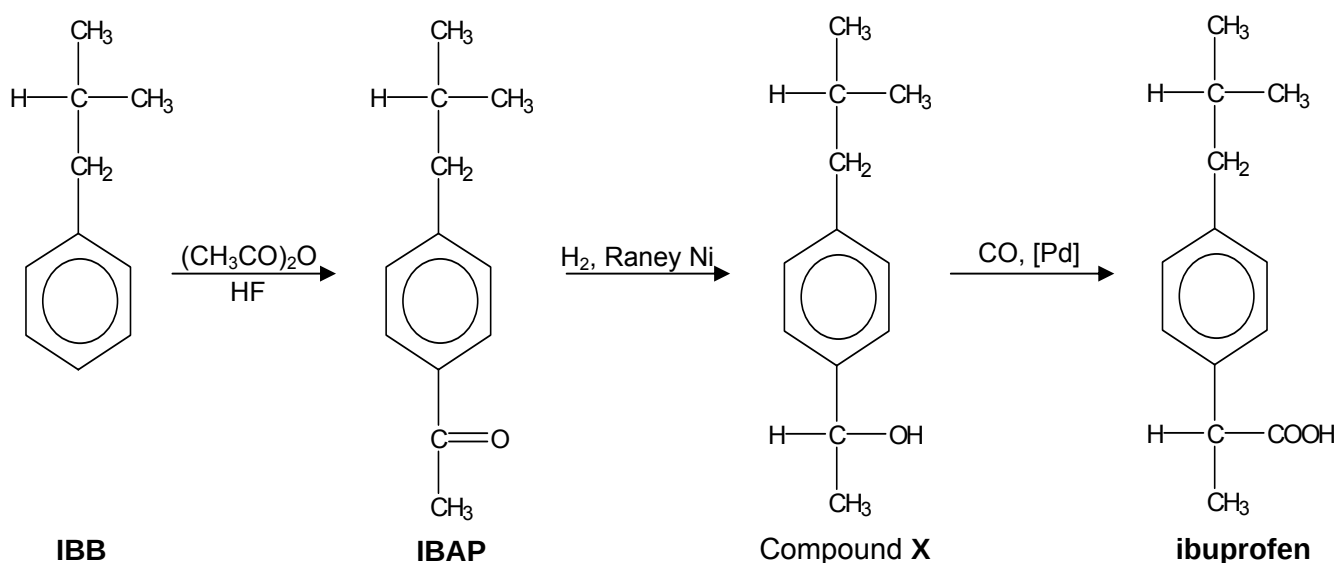
3. Ibuprofen, a non-steroid drug, was discovered by Boots Pharmaceuticals in the late 1960s and is used to treat arthritis and relieve pain, fever and swelling. It is now the third largest selling analgesic after paracetamol and aspirin.



ibuprofen

When ibuprofen was approved for over-the-counter (OTC) use in the United States in May 1984, it was the first new OTC pain relief medication to enter the marketplace in a generation. Ibuprofen is available OTC in 200-mg and 400-mg doses. The recommended dosage varies with body mass and indication. 1200 mg is considered the maximum daily adult dosage for OTC use.

The synthesis of ibuprofen consists of three steps as shown below.



- (a) A man bought some ibuprofen tablets of dosage 200 mg over the counter. He consumed one pill 4 times a day. Is this level of consumption safe for the man? Explain your answer, showing relevant calculations.

3. (b) Name the functional group, other than the phenyl group, present in **IBAP**.

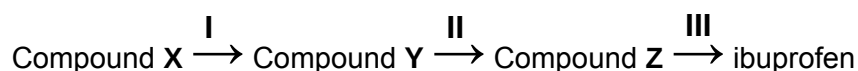
..... [1]

- (c) Describe a simple chemical test to distinguish between **IBB** and **IBAP**, giving the reagents and conditions required and observations expected for each.

.....

[2]

- (d) Suggest reagents and conditions that can be used to convert Compound **X** to Ibuprofen in a 3-step synthesis route in the laboratory. Compounds **Y** and **Z** are intermediates.



Step	Reagents and conditions
I	
II	
III	

[3]

- (e) State and explain the relative acidity of Compound **X** and Ibuprofen.

.....

[2]

3. (f) A student is investigating the percentage purity of ibuprofen in a sample of tablets. 6.00 g of a sample of impure ibuprofen tablets was crushed and dissolved in 200 cm³ of 0.200 mol dm⁻³ NaOH(aq). 25.0 cm³ of this solution was withdrawn and titrated against hydrochloric acid. The unreacted NaOH in this solution required 24.40 cm³ of 0.120 mol dm⁻³ of hydrochloric acid for complete neutralisation.

(i) Write a balanced equation for the reaction between sodium hydroxide and ibuprofen.

(ii) Calculate the amount of hydrochloric acid used in the titration.

(iii) Calculate the amount of sodium hydroxide left unreacted after dissolving the tablets.

(iv) Calculate the amount of sodium hydroxide reacted with the ibuprofen tablets.

3. (f) (v) Calculate the mass of pure ibuprofen in the tablets.

(vi) Calculate the percentage purity of ibuprofen in the sample.

[8]

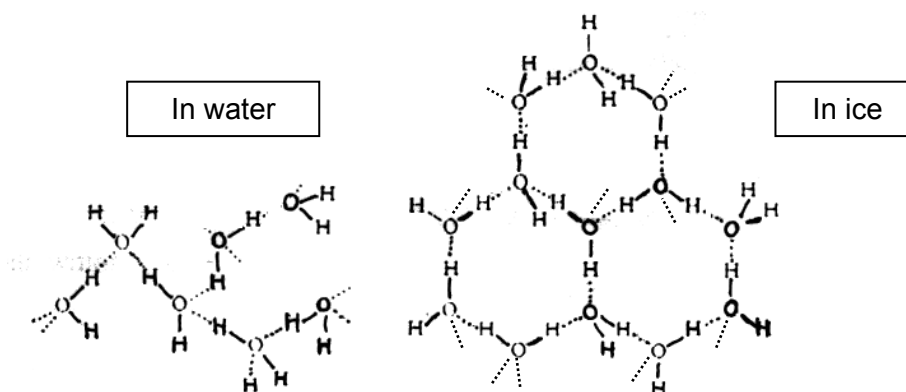
[Total: 17]

Section B

Answer **two** of the three questions in this section on separate answer paper.

4. Water is a liquid at ambient conditions, but it often co-exists on Earth with its solid state, ice, or gaseous state (water vapor or steam).

- (a) The diagram below shows the structure of water in the liquid and solid state.



- (i) Determine the type of intermolecular bonding between H_2O molecules.
- (ii) Water in the solid state is less dense than liquid water. Explain the difference in density in terms of their structures. [2]
- (b) Water in the liquid state is able to react with the chlorides and oxides of Period 3 elements, such as that of sodium, magnesium and aluminium.
- (i) The melting points of two chlorides are given below.

<i>Compound</i>	<i>formula</i>	<i>melting point / °C</i>
sodium chloride	NaCl	801
aluminium chloride	AlCl_3	178

Briefly relate these melting points to the structure and bonding in each of these chlorides.

- (ii) Describe the reaction of liquid water and chlorides of sodium, magnesium and aluminium, stating the approximate pH of any solution formed. Write equations where appropriate.
- (iii) The oxides of sodium, magnesium and aluminium also differ in their reactions with water.

Describe these reactions, relating them to the structure and bonding in each oxide. Write equations where appropriate.

[10]

4. (c) Water in the gaseous state is used as a chemical reagent in many reactions. One example is the water-gas shift reaction where water vapour reacts with carbon monoxide to form hydrogen and carbon dioxide.



- (i) Draw a dot-and-cross diagram to illustrate the bonding in CO_2 .
- (ii) Use the value of ΔH above and bond energies from the *Data Booklet* to calculate the total bond energy in the CO molecule.
- (iii) Suggest why the bond energy you have calculated in (c)(ii) is greater than either of the carbon-oxygen bond energies in the *Data Booklet*.
- (iv) At 773 K, 0.60 mol of steam and 0.40 mol of carbon monoxide are added into a 3-litre vessel. When equilibrium is reached, 0.20 mol of hydrogen is present in the mixture.

Calculate K_c , stating any units.

- (v) Predict with reasoning, the change, if any, on the equilibrium position for the reaction
 - when the volume is reduced.
 - when the equilibrium mixture is passed through NaOH(aq) .

[8]

[Total: 20]

5. (a) The reaction between aqueous bromine and methanoic acid is acid catalysed.



Student A performed an experiment to determine the order of reaction with respect to bromine. The initial concentration of bromine was $0.0100 \text{ mol dm}^{-3}$ and methanoic acid was used in large excess. The concentration of bromine at various times in the reaction mixture was found as follows:

Time / s	$[\text{Br}_2] / \text{mol dm}^{-3}$
0	0.0100
30	0.0090
60	0.0081
90	0.0073
120	0.0066
180	0.0053
240	0.0044
360	0.0029
480	0.0019
600	0.0013

- (i) Plot a graph of concentration of bromine against time and hence determine the order of reaction with respect to bromine.
- (ii) Why is methanoic acid used in large excess in this experiment? [4]
- (b) Student B performed the same experiment using different concentration of bromine and methanoic acid. The following results were obtained.

Experiment	Initial concentrations / mol dm^{-3}		Initial rate of formation of $\text{CO}_2(\text{g})$ / $\text{mol dm}^{-3} \text{ s}^{-1}$
	$[\text{Br}_2]$	$[\text{HCOOH}]$	
1	0.0100	0.0200	2.00×10^{-6}
2	0.0200	0.0400	8.00×10^{-6}

- (i) Deduce the order of the reaction with respect to HCOOH. Hence write a rate equation for the reaction.
- (ii) Hence calculate the rate constant for the reaction, stating its units. [3]

5. (c) The enthalpy change of neutralisation for ethanoic acid can be calculated from the temperature rise that occurs when aqueous ethanoic acid is neutralised by aqueous sodium hydroxide. In an experiment, 50.0 cm^3 of 1.30 mol dm^{-3} NaOH was placed in a polystyrene cup and 50.0 cm^3 of 1.20 mol dm^{-3} ethanoic acid was placed in a measuring cylinder. Initial temperatures of NaOH and CH_3COOH were recorded.

The acid was added to the alkali and the mixture was then stirred gently with the thermometer. The highest temperature reached was recorded.

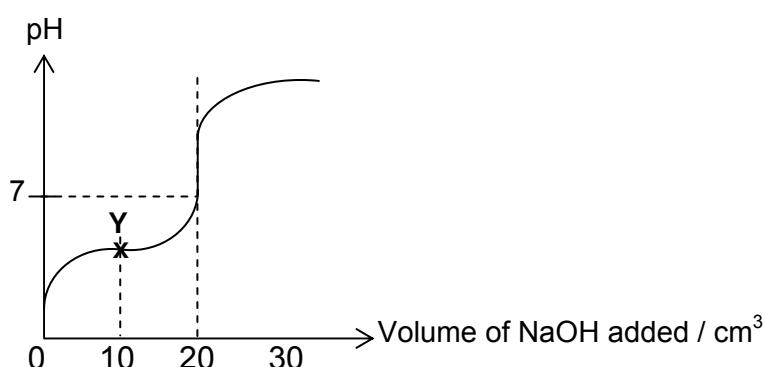
Results:

Initial temperature of NaOH / $^{\circ}\text{C}$	29.0
Initial temperature of CH_3COOH / $^{\circ}\text{C}$	29.2
Highest temperature reached / $^{\circ}\text{C}$	37.0

You should assume 4.2 J is required to raise the temperature of 1 cm^3 of solution by 1°C .

- From the experimental data, calculate the heat evolved in the reaction.
 - Using your answer to (c)(i), calculate the enthalpy change of neutralisation of CH_3COOH with NaOH.
 - When the experiment in (c) is repeated with HCl instead of CH_3COOH , the enthalpy change of neutralisation obtained is more exothermic. Suggest a reason for this observation.
- (d) In a separate experiment, 0.10 mol dm^{-3} of sodium hydroxide is slowly added to 20 cm^3 of ethanoic acid. The following titration curve was obtained.

[4]



- With the aid of the titration curve, determine the initial concentration of CH_3COOH .
- State and explain the choice of indicator for the above titration from the following list.

indicator	pH at which colour changes
methyl violet	0–1
methyl orange	3–4
bromothymol blue	6–7
phenolphthalein	9–10

5. (d) (iii) At point Y, the solution contains a mixture of CH_3COOH and CH_3COONa which resists pH changes when small amount of acid or base is added to it.

Using balanced equations, illustrate how the solution at point Y resists pH changes when small amount of acid or base is separately added to it.

[5]

- (e) How would you expect the K_a of chloroethanoic acid to compare with that of ethanoic acid. Explain your answer.

[2]

- (f) In the vapour state, the M_r of ethanoic acid was found to be 120. However, when ethanoic acid was dissolved in water, its M_r was 60. Using a suitable diagram, explain this behaviour exhibited by ethanoic acid.

[2]

[Total: 20]

6. (a) Ester **A** is used in synthetic fruit flavours. When **A** is subjected to acid hydrolysis, monoprotic acid **B** and **C**, C_3H_8O , are produced.

1.00 g of the monoprotic acid **B** requires 0.0165 mol of NaOH for complete neutralisation. **C** reacts with alkaline aqueous iodine giving a precipitate **D**.

- (i) Suggest suitable reagent and conditions for the hydrolysis of **A**.
- (ii) Calculate the M_r of acid **B**, and hence identify **B**.
- (iii) Deduce the structural formulae of **C** and **D**. State the type of reaction that occurs when **C** reacts with alkaline aqueous iodine.
- (iv) Hence identify the ester **A** and write a balanced equation for the acid hydrolysis of **A**.

[8]

- (b) An organic compound **E** has the following composition by mass:

C, 64.9%; H, 13.5%; O, 21.6%

E forms **F** when reacted with PCl_5 . When **F** is refluxed with ethanolic potassium hydroxide, three isomeric products **G**, **H** and **J** are produced. When heated with acidified $KMnO_4$, both **G** and **H** forms ethanoic acid only while **J** forms propanoic acid and a gas that forms white precipitate with limewater.

- (i) Calculate the empirical formula of **E**.
- (ii) Deduce the structural formulae of **E**, **F**, **G**, **H** and **J**. In your answer, include the type of reaction occurred and functional group deduced.
- (iii) State the type of isomerism exhibited by **G** and **H**.
- (iv) Write a balanced equation for the reaction of **H** with acidified $KMnO_4$.

[12]

[Total: 20]

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